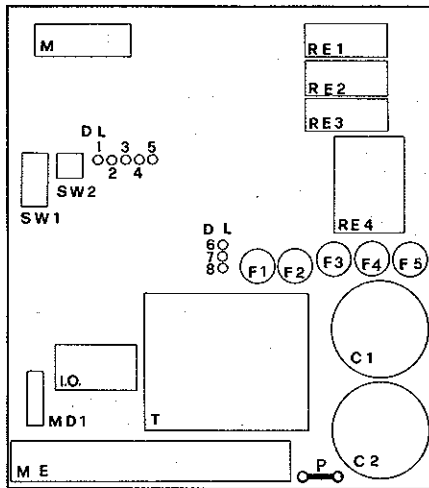


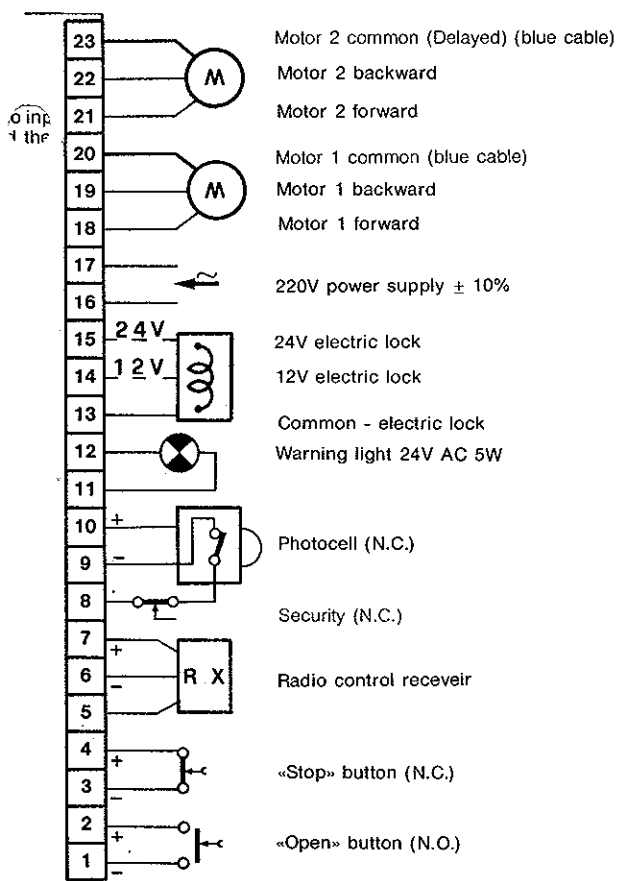
# 400 MP

10/85



- DL1 = LED diagnostic.
- DL2 = LED power relay.
- DL3 = LED open/close relay.
- DL4 = LED leaf delay relay.
- DL5 = LED electric lock.
- DL6 = LED «stop» button.
- DL7 = LED photocell.
- DL8 = LED «open» button.
- M = FAAC microprocessor.
- SW1 = Dip Switch assembly (open/close time-pause-reversing stroke selection).
- SW2 = Dip Switch assembly (logic selection-leaf delay-diagnostics).
- T = transformer.
- RE4 = motor relay.
- RE3 = direction relay.
- RE2 = leaf delay relay.
- RE1 = electric lock relay.
- ME = pull-out terminal strip.
- C1 = 16MF 250V capacitor.
- C2 = 16MF 250V capacitor.
- P = capacity selection bridge.
- F1 = 1A fuse-radio control and photocell.
- F2 = 1A fuse-warning light.
- F3 = 2A fuse-electric lock.
- F4 = 1A fuse-electronic circuit power supply.
- F5 = 5A fuse-motor power supply.
- MD1 = connector for board MD1.
- I.O. = pre-setting for «automatic input» system.

## TERMINAL STRIP WIRING



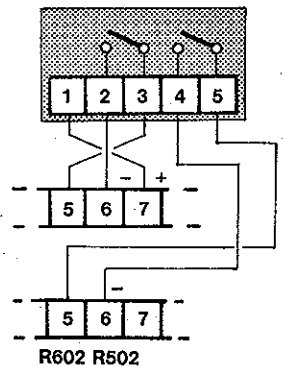
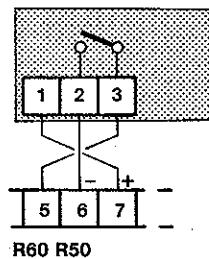
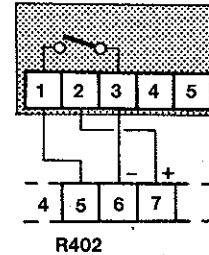
## WIRING:

The FAAC 400MP has a fully extractable terminal strip.  
Power supply: 220V ± 10%, 50/60 Hz, single phase; in the case of 220V three phase line, use 2 phases; in the case of 380V three-phase line, use one phase and neutral.

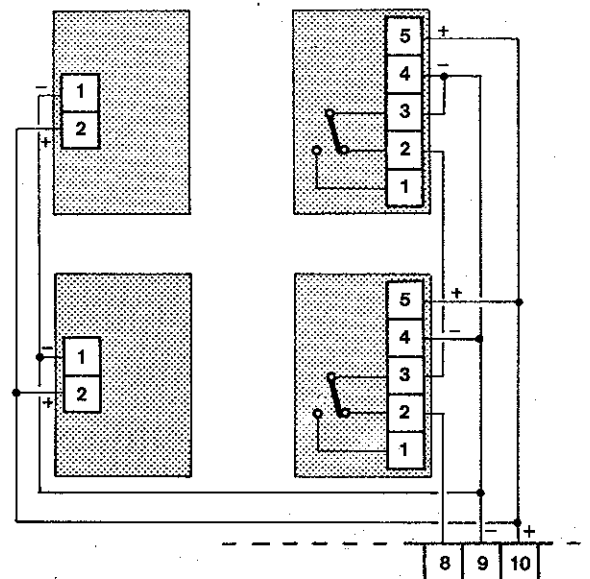
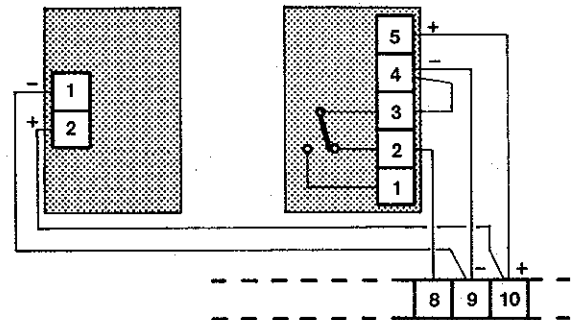
Use separate cables for power connections (power supply-motor connection and for control connections (buttons, radio control, photocells, etc. ...).

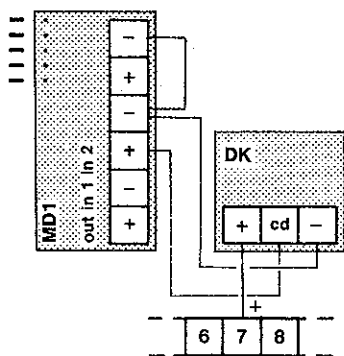
**Warning:** The power supply line should always be fitted with a 16A one-phase safety switch complete with 4A fuses or automatic 6A switch.

## Connection of the radio receiver control

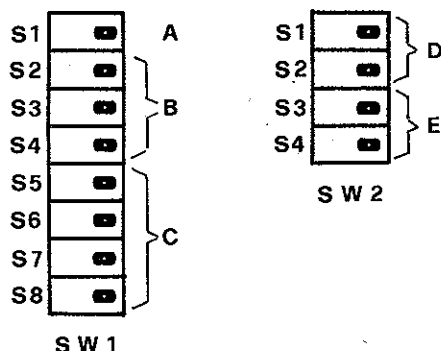


## Photocell connection





## PROGRAMMING THE FAAC 400MP SYSTEM



- A = Reversing stroke  
B = Pause time selection  
C = Open/close time selection  
D = Logic selection  
E = Diagnostics and leaf delay time selection

**Attention!** The electrical supply has to be taken off every time you select the planning dip-switches.

| SW1                          |     |     |     |      |  | SW2                     |     |     |      |                    |     |       |
|------------------------------|-----|-----|-----|------|--|-------------------------|-----|-----|------|--------------------|-----|-------|
| OPEN/CLOSE<br>TIME SELECTION |     |     |     |      |  | PAUSE<br>TIME SELECTION |     |     |      | LOGIC<br>SELECTION |     |       |
| S5                           | S6  | S7  | S8  | SEC. |  | S2                      | S3  | S4  | SEC. | S1                 | S2  | LOGIC |
| ON                           | ON  | ON  | ON  | 2    |  | ON                      | ON  | ON  | 2    | ON                 | ON  | A     |
| OFF                          | ON  | ON  | ON  | 4    |  | ON                      | ON  | OFF | 5    | OFF                | OFF | E     |
| ON                           | OFF | ON  | ON  | 6    |  | ON                      | OFF | ON  | 10   | OFF                | ON  | S     |
| OFF                          | OFF | ON  | ON  | 8    |  | ON                      | OFF | OFF | 15   | ON                 | OFF | M.P.  |
| ON                           | ON  | OFF | ON  | 10   |  | OFF                     | ON  | ON  | 20   | SW2                |     |       |
| OFF                          | ON  | OFF | ON  | 15   |  | OFF                     | ON  | OFF | 30   |                    |     |       |
| ON                           | OFF | OFF | ON  | 20   |  | OFF                     | OFF | ON  | 60   |                    |     |       |
| OFF                          | OFF | OFF | ON  | 25   |  | OFF                     | OFF | OFF | 120  |                    |     |       |
| ON                           | ON  | ON  | OFF | 30   |  | SW2                     |     |     |      |                    |     |       |
| OFF                          | ON  | ON  | OFF | 35   |  |                         |     |     |      |                    |     |       |
| ON                           | OFF | ON  | OFF | 40   |  |                         |     |     |      |                    |     |       |
| OFF                          | OFF | ON  | OFF | 45   |  |                         |     |     |      |                    |     |       |
| ON                           | ON  | OFF | OFF | 50   |  | SW2                     |     |     |      |                    |     |       |
| OFF                          | ON  | OFF | OFF | 60   |  |                         |     |     |      |                    |     |       |
| ON                           | OFF | OFF | OFF | 120  |  |                         |     |     |      |                    |     |       |
| OFF                          | OFF | OFF | OFF | 240  |  |                         |     |     |      |                    |     |       |

| SW1                 |     |  |  | SW2        |     |      |  |
|---------------------|-----|--|--|------------|-----|------|--|
| REVERSING<br>STROKE |     |  |  | LEAF DELAY |     |      |  |
| S1                  |     |  |  | S3         | S4  | SEC. |  |
| ON                  | YES |  |  | ON         | ON  | 0    |  |
| OFF                 | NO  |  |  | OFF        | ON  | 3,5  |  |
|                     |     |  |  | ON         | OFF | 7    |  |

| SW2         |     |     |  |
|-------------|-----|-----|--|
| DIAGNOSTICS |     |     |  |
| S3          | S4  |     |  |
| OFF         | OFF | YES |  |

## PROGRAMMING:

The FAAC 400MP has two Dip switch assemblies, SW1 and SW2, with which the whole system can be programmed.

Therefore:  
operating mode selection  
open/close time selection  
pause selection  
leaf delay selection  
«reversing stroke» selection  
«diagnostic» selection  
are obtained by setting the Dip Switches.

## OPERATING LOGICS

The FAAC 400MP system has four operating logics:

- mode M.P.: «set-up»  
mode A : «automatic»  
mode E : «semiautomatic»  
mode S : «safety»

## MP mode - «set-up»

The «set-up» logic works only opening; the «open» relay energizes when the button is pushed and de-energizes when the button is released.  
The «set-up» logic is used during installation to select the correct direction of rotation of the motor.

## A - «automatic»

When an «open» signal is sent, the gate opens up to limit switch, pauses for the preselected length of time and then closes automatically.  
If the «open» signal is sent while the gate is opening, it is disregarded and the gate simply goes on opening.  
If the «open» signal is sent during pause, pause time is reset and starts again from zero.  
If the signal is sent while the gate is closing, the gate «reverses» and opens again.

## E - «semiautomatic»

Using only one button or the radio control, you have the following working:  
the first signal commands «open»  
the second signal commands «stop»  
the third signal commands «close».  
If the signal is sent while the gate is closing, the gate «reverses».  
If the signal is sent when the gate is already open, and time expired, the gate closes.

## S - «safety»

The S works in the same way as the A except that, if the signal is sent while opening, the gate closes immediately without completing the cycle.

## Open-close time selection

The Dip Switches can be used to select 16 different open-close times: from a minimum of 2 seconds to a maximum of 240 seconds.

## Pause time selection:

With the same procedure, it is possible to select 8 different pause times; from a minimum of 2 seconds to a maximum of 120 seconds.

## GATE LEAF DELAY:

The FAAC 400MP has a built-in leaf time delay unit which can be used to delay the closing of one of the two leaves.  
The leaf can be delayed by 3.5 or 7 seconds.

## REVERSING STROKE:

By setting Dip Switches S1 assembly, the systems can be set to «reversing stroke».

## OPERATION:

If an «open» signal is sent while the gate is closed, the system sends a brief «close» signal, energizes the electric lock and then sends an «open» signal.  
This is very useful whenever the electric lock is difficult to release.

## DIAGNOSTIC

The FAAC 400MP has an automatic trouble finding programme called «diagnostic programme». To use this programme, set Dip Switches S3 and S4 of the SW2 assembly. For use, refer to the relevant chapter.

## OPERATION OF SYSTEM:

### «Open» button

If the system is on and a short-circuit occurs on the «open» button or on the radio control receiver, the electronic control operates opening-closing and stop positions.  
This occurs in all 4 logics of operation and serves to ensure the gate stays shut even in the event of a failure.

### Stop button

The stop button works with N.C. contact.  
The pression of the button stops the gate in any operation phase. If another signal is sent after the STOP button has been pressed, the FAAC MP400 gives a 5 seconds warning (flashing warning light) and then closes the gate.

### Photocell

The photocell remains on all the time and works with three wires: two for 30 D.C. power supply and one for pulse.  
The relay contact is normally closed.

### Operation:

When the gate is closing, the photocell operates the reversing of the gate.  
When the gate is opening, the photocell signal is disregarded.  
During gate pause, the photocell cancels pause time, gives a 5 seconds warning (flashing warning light) and then closes the gate.

### Warning light

The FAAC 400MP has only one warning light output:  
Voltage 24V; Power: 5W. Max.

### Operation:

LIGHT OFF: Gate completely closed.  
LIGHT FLASHING: Closing warning (5 seconds) - closed.  
LIGHT ON: opening-open-half open.

## ELECTRICK LOCK

The FAAC 400MP has both 12 V and 24 V outputs.  
The 24 V output makes it possible to connect two 12V electric locks in series.

## BRIDGE: (capacitors)

The system can control two motors. If only one motor is connected, bridge P must be cut so as to eliminate one of the capacitors.

## Setting card MD1

Card MD1 can be plugged straight into the FAAC 400MP.  
Therefore, you have to connect the Digi-key (THREE WIRES) with the card because all the internal connections are achieved by means of the plug.

## «Automatic input» device (Optional)

If you want to operate gate opening automatically: by means of photocell, induction loop, contact threshold, etc., an electronic device called «automatic input» available on request, must be fitted.  
The device works only when the system is in the A or S logic (automatic logics).

The electronic device has a extractable terminal strip for wiring and a warning LED.

The device can be inserted and welded in the special holes made in the printed base.

#### Operation:

When the vehicle passes over the automatic signal generator, the electronic control operates gate opening.

If the vehicle stays on the signal generator, the gate is kept open in pause. When the vehicle moves off, the system counts pause time and then closes the gate.

#### Led's

FAAC 400MP system is provided with LED'S serving to check system operation. The LED'S are divided into two main groups:

the first is used to check operation of inputs («open» button - «stop» button - photocells etc.)

the second group is used to check power supply of the various control relay coils on the system (motor relay-direction relay-leaf delay relay-electric lock relay). There is also a «diagnostic led» to display all the automatic tests carried out by the system.

#### Operation:

##### INPUT LED'S:

The LED'S are lit when the contact on the terminal strip is closed.

Therefore, the inputs with N.C. contact will have LED on and goes off when the input is busy.

The LED of inputs with N.O. contact, on the other hand, is normally off and goes on when the input is busy.

##### Relay led's

The relay LED'S light up when the associated relay coils are on.

The above two groups of LED'S make it easy to check operation of the system and of the accessories connected to it.

#### DIAGNOSTIC

The FAAC 400MP systems have an automatic trouble finding programme, called «diagnostic programme».

To set the diagnostic programme, proceed as follows:

- 1) Turn off the power supply
- 2) Remove the motor fuse
- 3) Set Dip Switches S3 and S4 of the SW2 assembly to OFF
- 4) Turn on power supply again.

#### Operation

With the diagnostic programme set, the following tests are carried out:

- 1) «Microprocessor FAAC test».

This test does not appear because it is carried out inside the microprocessor and lasts just a few msec.

If FAAC microprocessor is working, the «diagnostic led» flashes for 5 seconds. If FAAC microprocessor is not working, the «diagnostic led» stays on or off and the system stops the test cycle.

- 2) Internal tests

Automatic sequence:

- a) Motor power supply relay test.
- b) Direction relay test.
- c) Leaf delay test.
- d) Warning light test.
- e) Brake or electric lock test.

Each relay has a LED associated to it which lights up when the relay coil is on. To check whether the relays are working, simply check the sequence on the Led's and their energization.

After this stage, which lasts 30 seconds, the diagnostic LED flashes for 5 seconds and then stays on, presetting the system for the «input circuit» and Dip switches tests.

If no input has been tested after 50 seconds, the microprocessor test is repeated and the «diagnostic» cycle re-started.

- 3) Input circuits and dip switch tests

- a) Input circuits

The microprocessor reads and stores the input configuration.

If the configuration changes, it warns the user by flashing the «diagnostic led». The inputs can be tested in any order.

Example:

«open button operating test»

Push and then release the button. Watch the LED associated to the input circuit. If the signal is received, the «diagnostic led» flashes for about 5 seconds.

If the signal is not received the «diagnostic led» stays on, thus warning the user that the button is not working.

All the accessories connected to the sustem can be tested in this way.

- b) Dip switch test

The procedure followed for the input circuits can be used to check the operation of the dip switches as well.

To check dip switch S1 (set at OFF), simply set it to ON for an instant (if the switch is set at ON, turn if OFF for an instant).

If the signal is received, the «diagnostic led» flashes for about 5 seconds.

If the signal is not received, the «diagnostic led» stays on, thus warning the user that the dip switch is not working.

All the dip switches can be tested in this way.

N.B. If the system is set to «diagnostic» while the «open» button or the radio control receiver is short circuited, the automatic «internal tests» cycle will not be carried out and the diagnostic LED stays on instead of flashing.

To exit from the trouble finding cycle, proceed as follows:

- 1) Turn off the power supply.
- 2) Change position of dip switches S3 and S4 for the SW2 assembly.
- 3) Put the motor fuse back in.
- 4) Turn on power supply again.